Daya Lokesh Duddupudi

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PROJECTS

Distributed Road Capacity Booking System,
Python, FastAPI, PostgreSQL, PostGIS, SQLAlchemy
- Engineered distributed microservices architecture with central coordinator and regional managers, enabling scalable road capacity booking across multiple geographical regions with 99.5% consistency
- Implemented two-phase commit protocol (reserve/confirm) ensuring ACID properties for cross-region transactions, handling 1,000+ concurrent booking requests with zero data corruption
- Integrated geospatial database (PostgreSQL/PostGIS) with GeoAlchemy2 and Shapely, optimising road segment queries by 60% through spatial indexing and reducing average query time to under 200ms
- Deployed Open Source Routing Machine (OSRM) API for dynamic route calculations, processing real-world road networks with 150,000+ segments and supporting multi-region route planning

Cloud-Architected IoT System for Wildfire Detection,
C++, ESP32, LoRa, Azure, FastAPI
- Prototyped distributed IoT network with 8 heterogeneous edge devices for real-time wildfire detection, integrating temperature (AM2302), smoke (MQ-2) sensors, and ESP32-EYE cameras for visual verification
- Developed embedded firmware in C++/Arduino for ESP32 microcontrollers, implementing dual-communication strategy using Wi-Fi and LoRa mesh network, achieving 2km+ range in remote areas
- Architected event-driven Azure backend with IoT Hub, Event Hubs, and Functions for scalable data ingestion, designed to process 10,000+ sensor readings per minute with sub-second latency
- Designed automated alerting pipeline using Azure Notification Hubs, enabling real-time emergency response triggers and integration with suppression systems for wildfire management

LEO Satellite Network Simulation for Offshore Wind Telemetry, *Python, Flask, RSA, NumPy, Leaflet.js*
- Simulated Low Earth Orbit satellite constellation with 12 satellites relaying offshore wind farm data, modelling realistic propagation delays, signal noise, and network topology changes
- Implemented dynamic routing algorithm using weighted Dijkstra's approach, optimising data paths based on real-time link quality (SNR/BER) and reducing average latency by 35%
- Secured data transmission with 2048-bit RSA encryption and Hamming (7,4) forward error correction, achieving 99.8% data integrity across simulated 500km+ transmission distances
- Built Flask web dashboard with Leaflet.js and Chart.js visualisations, displaying real-time satellite positions, active routes, and turbine metrics for 50+ offshore wind turbines

Sustainable City Management Platform,
Django, PostgreSQL, Docker, AWS, Nginx, Celery, Redis
- Deployed full-stack platform on AWS EC2 with Docker containerisation and Nginx reverse proxy, implementing SSL/TLS protocols for end-to-end encryption and serving 500+ concurrent users
- Architected automated bus rerouting algorithm and RESTful API processing real-time congestion data, reducing average commute times by 18% across 200+ bus routes in simulation
- Engineered asynchronous data aggregation pipeline using Celery and Redis, processing 100,000+ transit data points hourly and reducing dashboard load times by 55%
- Developed interactive heatmap visualisations and trend analysis dashboard with Django and PostgreSQL, enabling data-driven decision-making for urban planning scenarios

PROFESSIONAL EXPERIENCE

National Institute of Technology Arunachal Pradesh Jote, India, *Research Intern*
03/2023 – 04/2023 | Jote, India
- Architected privacy-preserving machine learning framework using Federated Learning and Blockchain schemas, enabling verifiable model training across 50+ distributed nodes without exposing sensitive medical data
- Co-authored IEEE publication on Homomorphic Re-encryption techniques, reducing data exposure risk by 85% while maintaining 94% model accuracy for medical diagnosis applications

Anna University, *Research Intern*
12/2022 – 01/2023 | Chennai, India
- Developed Convolutional Neural Network (CNN) model achieving 89% accuracy in signature forgery detection, processing 5,000+ signature samples across multiple verification scenarios
- Optimised deep learning architecture through hyperparameter tuning and data augmentation, reducing false positive rate by 32% and improving model generalisation across diverse signature styles
- Implemented preprocessing pipeline using OpenCV and NumPy, accelerating training time by 40% through efficient image normalisation and feature extraction techniques

EDUCATION

Master of Science in Computer Science, Future Networked Systems, *Trinity College Dublin*
09/2024 – Present | Dublin, Ireland

Bachelors of Technology, Computer Science and Engineering, *National Institute of Technology, Arunachal Pradesh*
12/2020 – 05/2024 | Jote, India
- Graduated with 8.00/10.00 GPA

TECHNICAL SKILLS

LANGUAGES
Python, C/C++, SQL, JavaScript, HTML/CSS, Matlab

FRAMEWORKS
FastAPI, Django, Flask, React, TensorFlow, Keras, Scikit-learn

Developer Tools
Git, Docker, AWS, Azure, Google Cloud, Linux, Nginx, VS Code, PyCharm, N8N

LIBRARIES
NumPy, pandas, SQLAlchemy, GeoAlchemy2, Shapely, OpenCV, Matplotlib, Celery, Redis

PUBLICATIONS

Privacy-Preserving Sensitive Data on Medical Diagnosis using Federated Learning and Homomorphic Re-encryption, *IEEE*
06/2023

Privacy-Preserving and Verifiable Decentralised Federated Learning, *IEEE*
06/2023